Verification Status Report

RISCFORGE 1TOPS SoC

VERIF-STATUS-v4.0 | September 2026

Document Overview

This document captures the current verification status of the RISCFORGE 1TOPS SoC across two scopes: peripheral IPs and system-level integration. It is intended as a living tracker for the verification team.

Field	Value
SoC	RV32IM_Zicsr single-core MCU, 32 KB ISRAM + 32 KB DSRAM, 64B BootROM
Peripherals	I2C, SPI, UART, GPIO, Timer 0, Timer 1
Bus Fabric	AXI-Lite crossbar (2 SLV → 3 MST) + AXI-to-APB bridge (6 slaves)

Peripheral IP Verification Matrix

The table below summarises verification progress for each peripheral IP across four categories.

IP Block	Bus	Func. Verif.	Lint	UVM Verif.	Datapath Verif.
APB I2C	APB	Done	Done	Not Started	Done
APB Timer (0 & 1)	APB	Done	Done	Done	Done
APB SPI Master	APB	Done	Done	Done	Done
APB UART	APB	Done	Done	Not Started	Done
GPIO	APB	Done	Done	Not Started	Not Started
AXI-Lite Crossbar	AXI	Not Started	Not Started	Not Started	Not Started
AXI→APB Bridge	AXI/APB	Not Started	Not Started	Not Started	Not Started

Peripheral Verification Summary

- **I2C**: 28-check directed regression. Covers single-byte write/read, data patterns (00/FF/55/AA), direction switching, reset recovery, and START/STOP generation. Multi-byte, clock stretching, and arbitration-loss scenarios remain open.
- **Timer**: Functionally verified via the SoC boot-to-runtime testbench (Timer 0 exercised as MMIO round-trip target). UVM testbench completed with compare-match IRQ, prescaler, and enable/disable coverage.
- **SPI Master**: 14-test BFM suite covering register R/W, FIFO depth, SPI modes, loopback, and RISC-V instruction injection. UVM environment completed. Also verified end-to-end through the SoC boot flow with both a lightweight BFM and the Infineon S25FS512S vendor model.

- **UART**: 16550-compatible. Functional testbench with 15 test cases + SVA written, simulation completed. TX/RX loopback, FIFO behaviour, and interrupt assertion covered.
- **GPIO, Crossbar, APB Bridge**: Not yet independently verified. The crossbar and bridge are exercised implicitly during SoC-level boot and runtime testing.

SoC-Level Integration Verification

Two full-SoC testbenches were developed, instantiating the complete `riscforge_soc_subsystem` with all IPs elaborated.

Boot Verification

Test Phase	Status	What Is Checked
Flash image integrity	**Done**	Image bytes read back correctly before boot starts
Cold boot completion	**Done**	`boot_done_o` asserts; all resets were low beforehand
Reset sequencing	**Done**	SPI released before core; core and peripherals released together
SRAM image integrity	**Done**	Every word in ISRAM matches the expected firmware image
Warm boot (soft reset)	**Done**	SRAM preserved, SPI fetch skipped, `warm_boot_q` set
Warm-boot retention	**Done**	Post-warm SRAM contents match pre-warm snapshot

Runtime Datapath Verification

An 11-instruction test program exercises three independent data paths after boot completes.

Data Path	Status	Description
Instruction fetch	**Done**	BootROM → ISRAM via `instr_fetch_demux`
Direct SRAM (data)	**Done**	Core → DSRAM via `hybrid_mem_interface` (1-cycle stall)
MMIO round-trip	**Done**	Core → AXI crossbar → APB bridge → Timer 0 and back

Note

The runtime test writes a control word to Timer 0's CTRL register over MMIO, reads it back, branches on the comparison result, and stores the pass/fail verdict into DSRAM over the direct path. This exercises all three data paths and proves that boot-loaded firmware executes correctly on the real core.

Outstanding Items

#	Item	Description
1	RISCOF compliance	Complete the rv32i, rv32m, Zicsr, Zifencei, and privilege test groups